TinyForestCD: A Lightweight Approach for Enhanced Detection of Forest Changes in Sentinel-2 Imagery

Changlong Lei, Enping Yan, Kaiqi Li, Dengkui Mo, and Jiawei Jiang

***Abstract*—Detecting forest changes through remote sensing imagery is essential for effectively managing and monitoring forest resources. Algorithms for forest change detection leveraging deep learning have shown remarkable performance across diverse scenarios. However, these algorithms face challenges such as imbalanced forest change samples and high false detection rates. To address these issues, this article proposes a global grouped coordinate attention (GGCA) mechanism, which improves the model's detection accuracy by emphasizing key features while suppressing irrelevant changes during the feature reconstruction process. Based on this, a lightweight forest change detection model, TinyForestCD, is designed. Furthermore, a hybrid loss function integrating CE loss and Dice loss is employed alongside an online hard example mining (OHEM) strategy, aiming to accurately extract key features from highly imbalanced forest change data. Extensive experiments conducted on a custom-built forest change detection dataset demonstrate that the TinyForestCD model outperforms other comparative models across various performance metrics, achieving a performance metric of 80.65% in *F*1-score and 67.57% in IoU. Moreover, in the practical deployment in Hunan Province, the average *F*1-score and IoU for accuracy assessment of six sample plots are 81.06% and 68.25%, respectively, which are consistent with the test set results. In addition, in the deployment results of the Shaoguan area, the model also showed the best performance compared with other models, validating the model's robustness and applicability in large-scale forest change detection. The research indicates that the TinyForestCD model has the potential for large-scale practical deployment, significantly improving the efficiency and accuracy of forest change surveys.**

***Index Terms*—Deep learning, forest change detection, global grouped coordinate attention (GGCA), Sentinel-2 images, Siamese network.**

I. Introduction

CHANGE detection through satellite observations has emerged as a fundamental research field, with emphasis on the analysis and measurement of temporal variations in terrestrial features using sequential imagery datasets [1]. Forests, as a vital component of the Earth's ecosystem, play an irreplaceable role in maintaining biodiversity, regulating hydrological cycles, and mitigating climate change. However, increasing human activities such as deforestation, urbanization, and infrastructure development significantly threaten the sustainability of global forest resources. Accurate and efficient forest change detection has thus become essential for governments and ecologists, providing crucial support for informed decision-making. In the above context, it is crucial to clarify the definitions of "forest" and "forest change." First, the FAO defines a forest as land with an area of more than 0.5 hectares and a tree canopy cover of over 10%. Based on this, in this study, a forest refers to Sentinel-2 covered pixels that fall within the FAO-defined range; deforestation is defined as the conversion of these pixels to nonforest categories during the period 2020–2022, confirmed through visual interpretation.

Early approaches to forest change detection relied primarily on manual visual interpretation [2]. While more efficient and cost-effective than traditional ground surveys, this method's heavy dependence on operator expertise often leads to misjudgments and omissions, reducing overall accuracy. To address this, researchers have progressively introduced methods based on algebraic approaches (e.g., change vector analysis [3] and vegetation index differencing [4]), image transformation techniques (e.g., principal component analysis [5]), and classification algorithms [6]. However, algebraic and image transformation methods are susceptible to atmospheric conditions, seasonal variations, sensor performance, and solar angles [7]. Similarly, classification-based algorithms are sensitive to background noise and prone to false detections in complex environments, limiting their adaptability to diverse scenarios.

Recent progress in computer vision has significantly promoted the extensive use of deep learning methods in image classification [8], object detection [9], and semantic segmentation [10]. As a specialized semantic segmentation task, remote sensing change detection has attracted growing interest. Daudt et al. [11] first introduced the Siamese network into convolutional neural networks (CNNs), proposing two variant models: FC-Siam-conc and FC-Siam-diff. Subsequent studies proposed models such as IFNet [12] with a fully convolutional dual-stream architecture, SNUNet-CD [13] with dense connections, and DSNUNet [14] based on dual Siamese branches.

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These models have achieved significant improvements in detection accuracy through various optimization strategies. However, existing methods require further advancements in capturing detailed features of change areas.

Transformer [15], known for its outstanding performance in natural language processing, has recently been introduced into change detection tasks. By utilizing the self-attention, the Transformer model long-range dependencies, outperforms traditional CNN methods. For instance, models like BIT [16], Changeformer [17], and AMTNet [18] excel in modeling long-range contextual relationships. However, the excessive computational overhead of the Transformer challenges its scalability in industrial applications. This has prompted the creation of lightweight models, including LSNet, which balances efficiency and accuracy by combining lightweight Siamese networks with feature pyramids (FPN) [19], and TinyCD, which achieves high detection accuracy by effectively utilizing low-level features within a Siamese U-Net framework [20].

Although these change detection methods have demonstrated strong performance with high-resolution images, the high cost and narrow coverage of such imagery restrict their practical application for large-scale deployment. Consequently, mid- to low-resolution remote sensing imagery has become a research focus for large-area forest change detection. For example, Kalinaki et al. [21] employed an improved U-Net model with Sentinel-2 data to detect forest changes in Brunei; de Bem et al. [22] used CNNs with Landsat imagery to monitor forest changes in the Amazon; and Xiang et al. [23] designed an optimized U-Net++ model with Sentinel-2 data to detect forest changes in Hunan Province, China. However, most existing methods overlook the critical challenge of extreme data imbalance in forest change detection. Although deforestation appears limited at the provincial scale, its scale and impact vary greatly across regions, with some areas experiencing severe and rapid forest loss. This uneven forest change poses a severe challenge to existing change detection models. Additionally, seasonal variations and textural similarities in forests increase the false detection rate. To surmount these barriers, this study introduces a lightweight model, TinyForestCD, built on the Siamese U-Net architecture and validates its performance using a forest change detection dataset. The results show that this method suppresses false detections in extremely class-imbalanced scenarios and shows significant advantages over cutting-edge methods on forest change detection datasets.

The key contributions of this study can be outlined as follows.

1) A novel forest change detection dataset has been created, consisting of 3221 image pairs with a resolution of 256 × 256 pixels, and featuring precisely annotated forest change regions through manual labeling.
2) A lightweight forest change detection model, TinyForest-CD, is proposed, incorporating the global grouped coordinate attention (GGCA) module to enhance target feature representation while suppressing background noise.
3) To address data imbalance, a hybrid loss function (CE loss and Dice loss) and online hard example mining (OHEM) strategy are introduced, significantly improving the detection performance of change regions.
4) A large number of comparative experiments have verified the excellent ability of TinyForestCD in suppressing false detections; large-scale practical deployments have verified the effectiveness of TinyForestCD.

The structure of this article is as follows. Section II outlines the materials and methods. Section III presents the experimental findings. Section IV discusses the results and explores future research possibilities. Section V summarizes the study's key contributions.

II. Materials and Methods

The main workflow of this study consists of four parts (Fig. 1): 1) collection of Sentinel-2 imagery data and forest change data; 2) processing of the collected data to construct a forest change detection dataset; 3) assessing the superiority of our proposed TinyForestCD model through three aspects: benchmarking against other advanced methods, ablation experiments, and comparison of replacing GGCA with other lightweight attention mechanisms; and 4) large-scale practical deployment and sampling accuracy assessment in Hunan Province using the TinyForestCD model to validate the model's effectiveness.

A. Study Area and Migration Verification Area

The study focuses on the Changsha, Zhuzhou, and Xiangtan regions in Hunan Province (Fig. 2), situated in its central and eastern areas. The geographical coordinates range from 26°03'N–28°40'N and 111°53'E–114°15'E, spanning a total area of roughly 28 000 square kilometers. The topography of the area is dominated by hills, and the climate is subtropical monsoon, nurturing a rich and diverse range of vegetation types and forest ecosystems, with a high forest cover rate. As this region has experienced rapid urbanization, it has had a certain impact on the surrounding forest resources, making it an important research area for forest change detection. In addition, to systematically verify the cross-regional transferability of the TinyForestCD model, this study constructed a dual verification system: 1) at the provincial scale, the entire Hunan Province (24°38'N–30°08'N, 108°47'E–114°15'E) was selected for intraprovincial transferability verification. The climate type of this region covers the transition zone from the mid-subtropical to the northern subtropical and 2) at the cross-provincial scale, Shaoguan City, Guangdong Province (23°53'N–25°31'N, 112°53'E–114°45'E) was selected as the verification area. Although Shaoguan borders Hunan Province and has similar terrain and vegetation types, it exhibits distinct climatic and ecological conditions due to the Nanling Mountain range, which serves as a significant climatic boundary in southern China. The Nanling Mountains create a transitional climatic zone between the central subtropical climate of Hunan and the southern subtropical climate of Shaoguan. As a result, Shaoguan experiences higher annual temperatures, increased precipitation, and a longer growing season,

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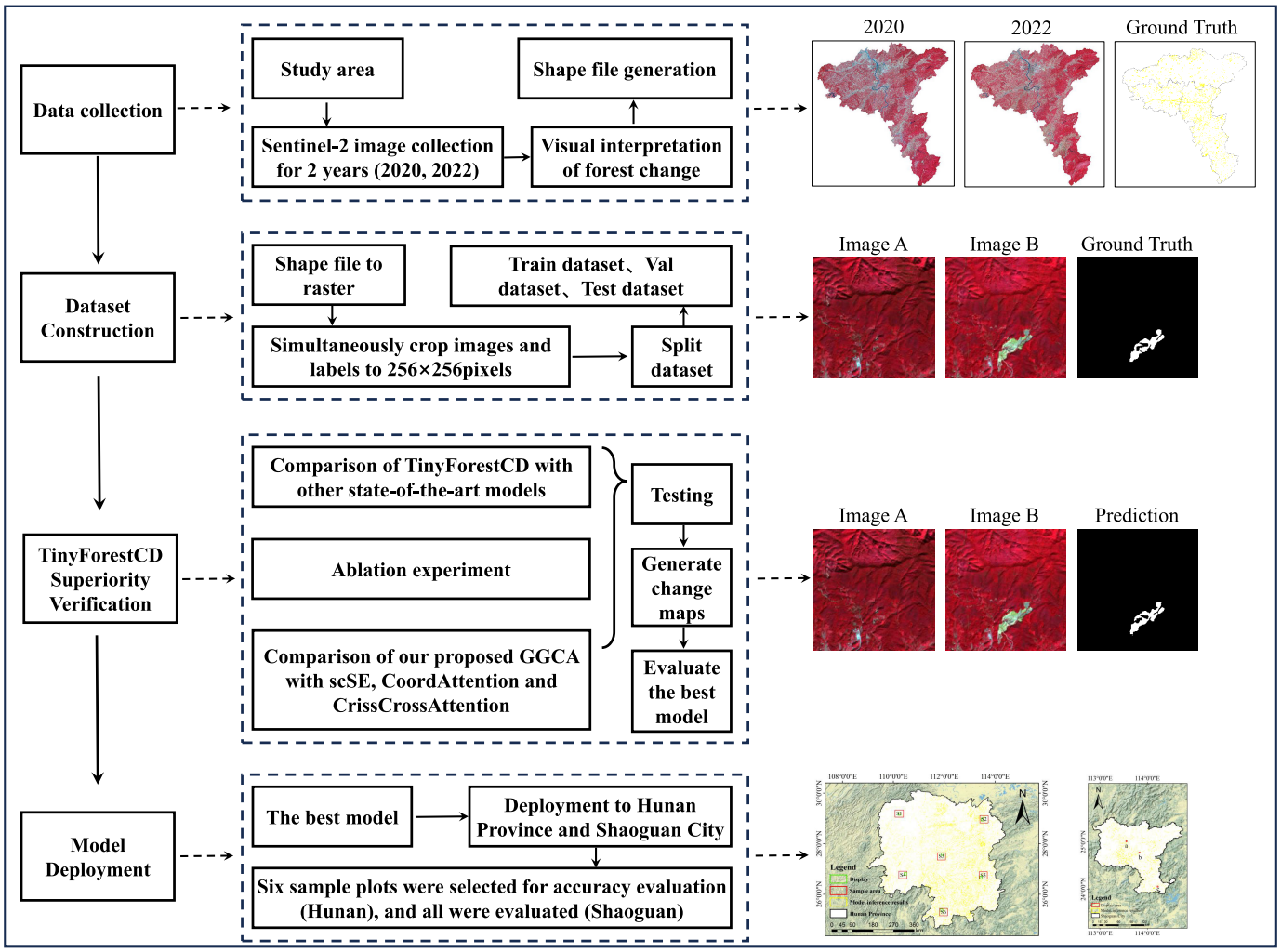


Fig. 1. Main research workflow.

leading to notable differences in vegetation composition and phenological characteristics compared to Hunan. The dominant forest types in Hunan are Pinus massoniana and mixed conifer-broadleaf forests, while Shaoguan is characterized by evergreen broadleaf forests and subtropical secondary vegetation. This cross-provincial multiregional verification idea can effectively test the adaptability of the model to the transition of climate zones.

B. Data Sources

The satellite imagery utilized in this research comprises Sentinel-2 Level-2A data with a spatial resolution of 10 m. Sentinel-2, characterized by a high revisit frequency of five days at the equator and a wide swath width of 290 km, is a highly suitable data source for forest change detection [24]. The optical imagery data originated from the open-access repository of Google Earth Engine [25]. These datasets have undergone orthorectification, radiometric correction, and atmospheric correction. Sentinel-2 data includes 13 spectral bands spanning visible light to shortwave infrared. To improve the performance of the model in minimizing the sensitivity to nonvegetated areas and considering the computational overhead, only three bands—near infrared (Band 8, rationale: distinguishes vegetation from nonvegetated surfaces), red (Band 4, rationale: key indicator for vegetation vigor and chlorophyll absorption), and green (Band 3, rationale: sensitive to vegetation reflectance in visible range)—were selected as input features for the model. To capture more representative samples of forest changes, images from two time intervals spanning two years were collected via the GEE platform. The imagery from autumn 2020 was used as the prechange period, while the imagery from autumn 2022 was used as the postchange period. These images were processed using a median composite approach, and cloud coverage was controlled to remain below 10%. Subsequently, forest change vector data were manually generated using visual interpretation in ArcMap 10.8. These vectors were created within forested regions defined by the "Woodland Resources Map" dataset [Fig. 2(b)] to guarantee the precision and dependability of the change data. In total, 7623 change vectors were identified and annotated. Comprehensive details about the data adopted in this study are presented in Table I.

C. Dataset Construction

The ground-truth vector data was rasterized to generate forest change raster data matching the Sentinel-2 imagery,

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TABLE I
DATA SOURCES

Data Series	Name of Data	Data Source	Band (Center Wavelength)	Spatial Resolution	Time
Remote sensing imagery	Sentinel-2 L2A	European Space Agency (ESA)	Band3(560nm) Band4(665nm) Band8(833nm)	10m	2020.9.1-2020.11.30; 2022.9.1-2022.11.30
Vector data	Forest change patch	Visual interpretation	/	/	/
	Woodland Resources Map	Academy of Forestry Inventory and Planning, State Forestry Administration, P.R. China	/	/	2020

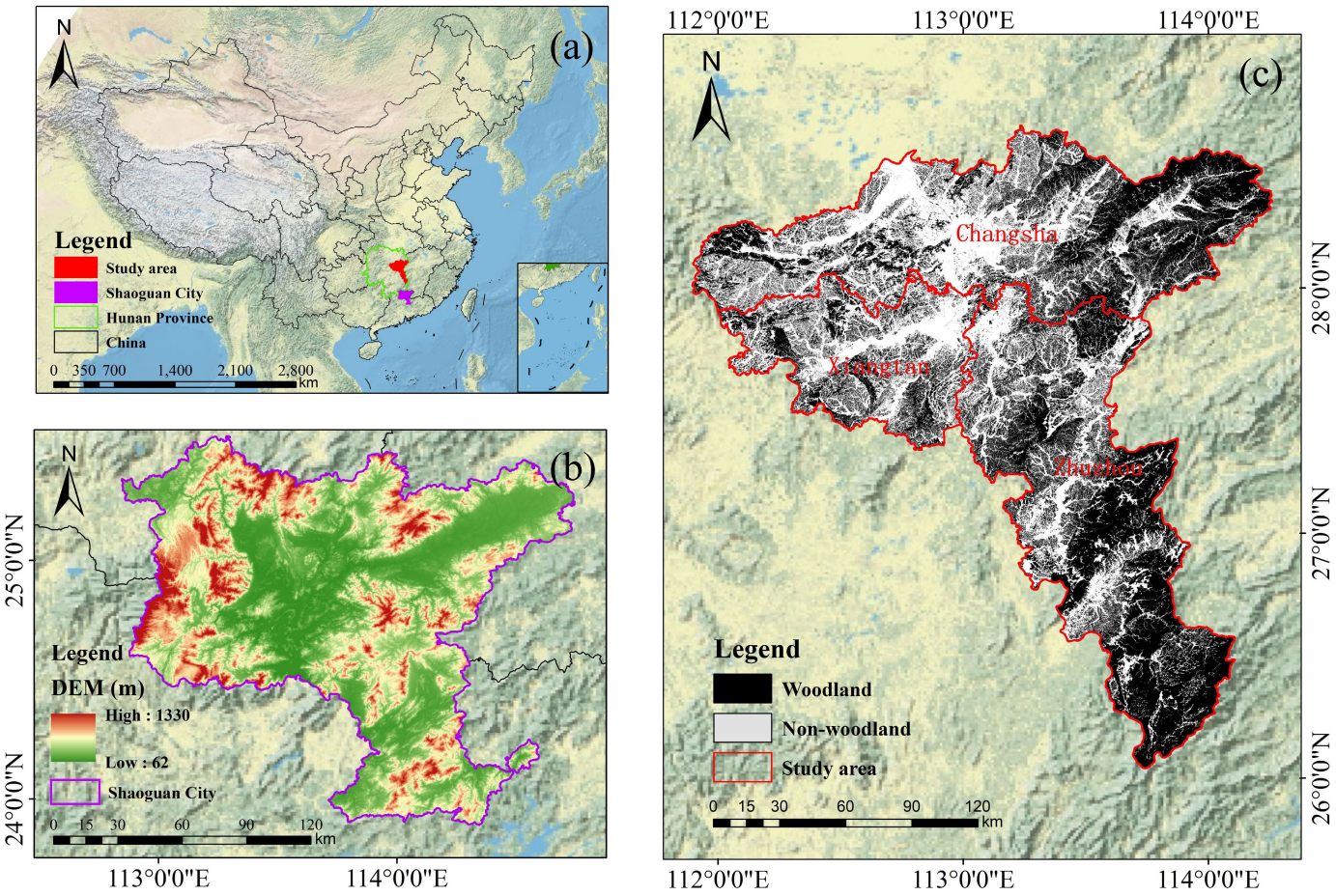


Fig. 2. (a) Geographical location of the study area and the verification area. (b) Elevation map of Shaoguan City. (c) Woodland Resources Map.

with a resolution consistent at 10 m. Subsequently, both the Sentinel-2 imagery and the forest change raster data were synchronously cropped to produce 256-pixel × 256-pixel image pairs with an overlap rate of 0.25. Given that most areas exhibited no changes, the dataset was characterized by a highly imbalanced ratio of positive to negative samples. To mitigate the impact of this imbalance, image pairs with no change information were manually excluded, resulting in a final dataset comprising 3221 image pairs. The dataset was partitioned into ten equal-sized, mutually exclusive subsets. In each iteration, nine folds were used for model training/validation, and the remaining fold was held out for testing. This process was repeated ten times so that each fold served once as the test set, and the final accuracy and other metrics were reported as the average across the ten runs, thereby reducing potential overfitting and improving generalization. Fig. 3 illustrates examples of Sentinel-2 imagery alongside their corresponding forest change data. Table II summarizes the positive sample percentage in the dataset, highlighting the significant imbalance between the two classes.

D. TinyForestCD Model Architecture

This article proposes a change detection model called TinyForestCD, with its architecture shown in Fig. 4. The model adopts a Siamese U-Net-like architecture design and consists of six key components: a Siamese encoder using the first three layers of EfficientNet-b4, a mix and attention mask block (MAMB), a mixing block, a decoder with three upsampling layers, a GGCA module, and a pixel-wise multilayer

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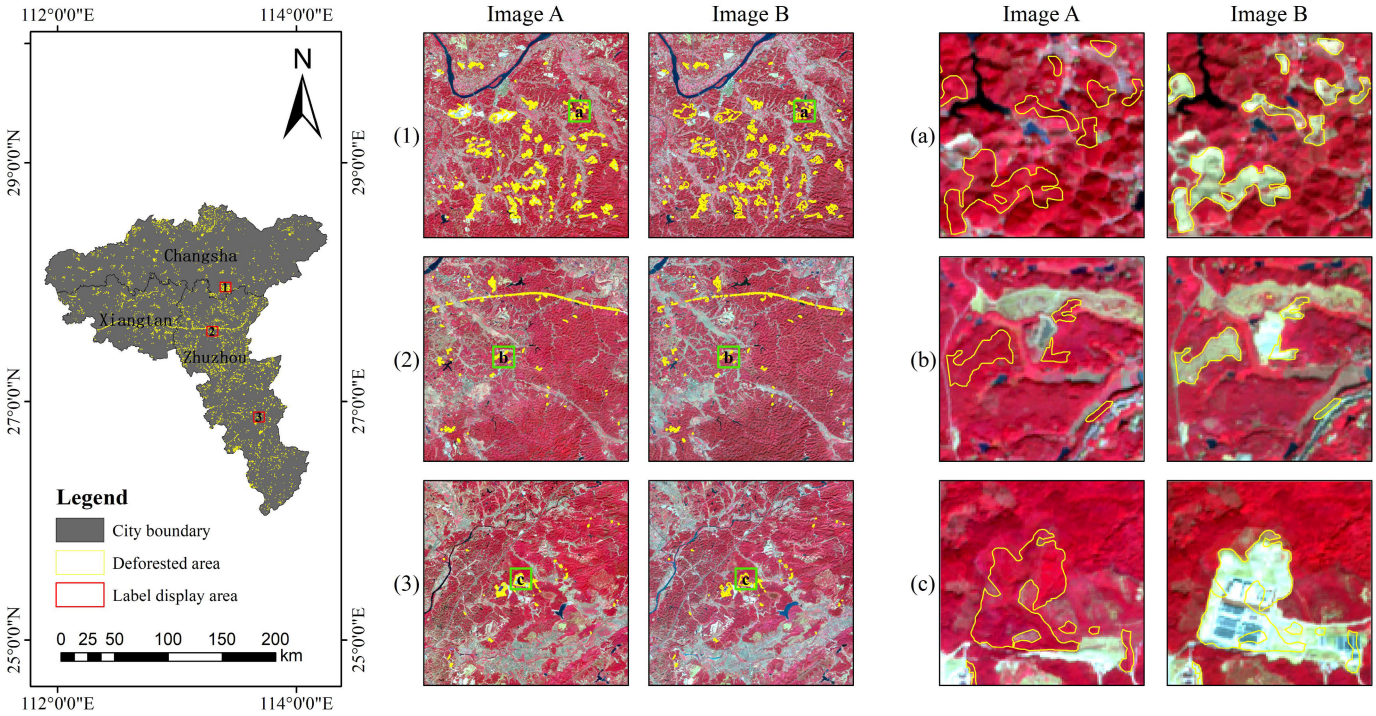


Fig. 3. Example of change labels. The yellow areas represent the regions of change. Images A and B were acquired in the fall of 2020 and 2022, respectively. (1), (2), and (3) are partially zoomed-in examples. (a), (b), and (c) are zoomed-in views of the change regions in (1), (2), and (3), respectively.

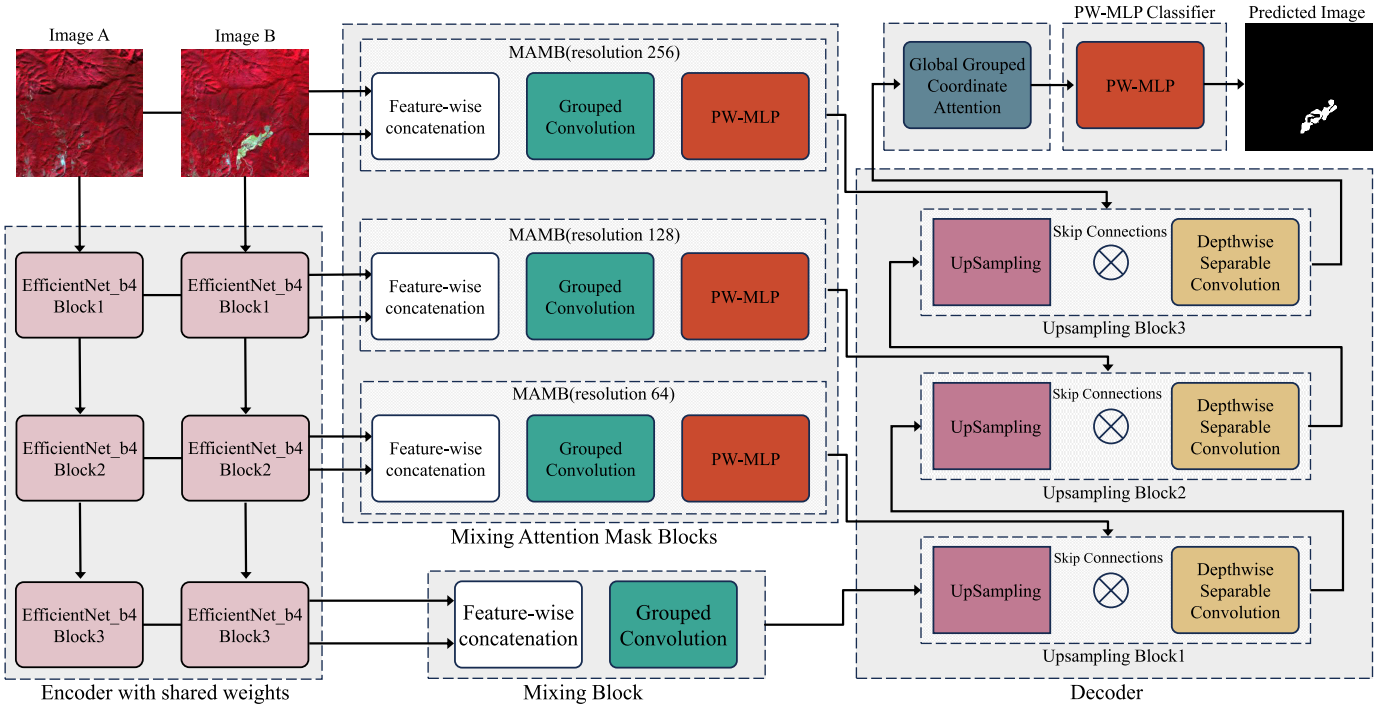


Fig. 4. Overall architecture of TinyForestCD.

TABLE II
PROPORTION OF POSITIVE SAMPLES IN THE DATASET

Dataset	Train Dataset	Val Dataset	Test Dataset	All Dataset
Number of samples	2576	322	323	3221
Percentage of positive sample area (%)	0.62	0.62	0.56	0.62

perceptron (PW-MLP). To more clearly explain the working principles of the model, this article uses $M \in \mathbb{R}^{C\times H\times W}$ to represent the reference tensor (Image A) and $N \in \mathbb{R}^{C\times H\times W}$ to represent the contrast tensor (Image B). C symbolizes the

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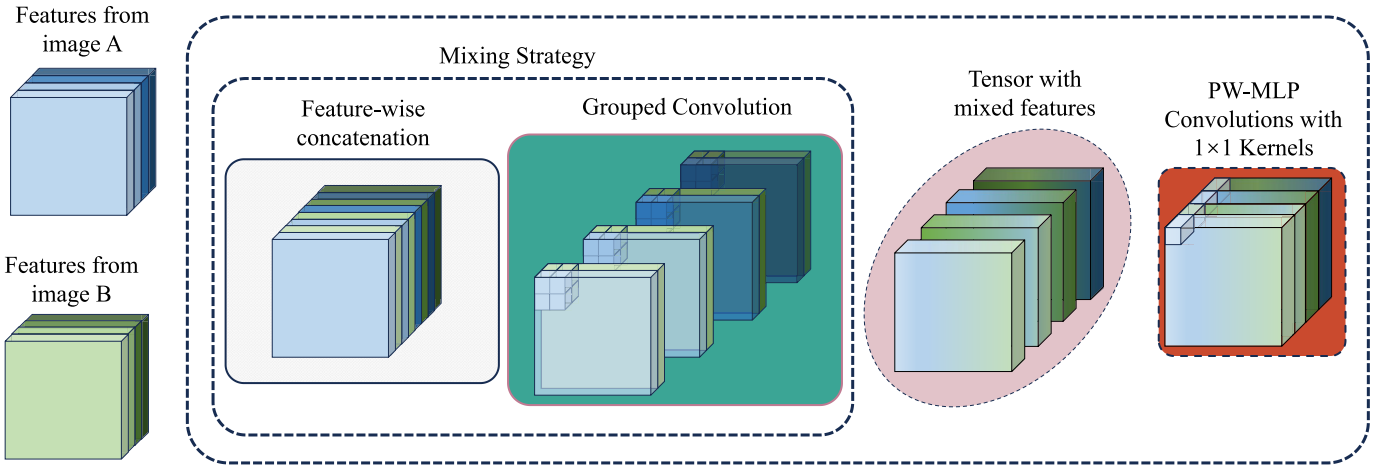


Fig. 5. Visual representation of the complete MAMB module.

number of channels, H signifies the tensor height, and W denotes the tensor width. Conv, PReLU, and IN2d represent the convolution operation, parametric rectified linear unit, and instance normalization, respectively.

1) Siamese Encoders: In the early stages of deep neural network training, the vanishing gradient problem often leads to training difficulties [26]. To address this issue, existing methods such as the skip connections in U-Net [27] have been widely adopted. However, training deep networks on small-scale datasets still faces significant challenges. Therefore, using pretrained backbone networks is a common and effective solution. This article selects the first three layers of EfficientNet_b4 as the backbone network, as it provides an optimal tradeoff between efficiency and performance [28].

The encoders, which share weights, are employed to extract image features from both the pretime and posttime phases. Assuming that the features extracted by the kth backbone block in EfficientNet_b4 are denoted as $M_k = f_k(M)$ and $N_k = f_k(N)$. The features extracted from each layer are not only used to compute the output but also to generate attention mask pairs (M_k, N_k). The features M_e and N_e produced by the last backbone block are input into the bottleneck mixing block to output a mixed tensor, which is used as the input to the decoder.

2) MAMB and Mixing Block: The MAMB fuses the features (M_k, N_k) and generates a mask Y_k, which is then applied to the skip connections to optimize the low-resolution information obtained in the upsampling stage. The MAMB consists of a mixing block and a pixel-wise mask generator PW-MLP (Fig. 5). In the MAMB, M_k and N_k possess the same shape (C_k, H_k, W_k) as well as an identical feature arrangement. Based on this, the network concatenates the tensors M_k and N_k in tensor Z_k according to the following definition:

$$Z_k^c = \begin{cases} M_k^{c/2}, & c \text{ even} \\ N_k^{(c-1)/2}, & c \text{ odd} \end{cases} \quad \forall c \in \{0, \ldots, 2C_k - 1\}. \tag{1}$$

Through group convolution, M_k and N_k can be mixed spatially and temporally. By defining the number of groups as C_k, C_k depth-2 kernels are obtained, which deal with the channels in tensor Z_k in pairs. This design ensures that each pair of corresponding channels (one from each input) is convolved within the same group. This enables channel-level correspondence between the two features, allowing the model to focus on fine-grained change patterns while maintaining the minimal parameter size. Moreover, by leveraging the correlation between semantically similar features, these kernels can perform spatial and temporal convolution simultaneously. The new tensor $Z'_k \in \mathbb{R}^{(C_k \times H_k \times W_k)}$ is defined as follows:

$$Z'_k = \text{PReLU}\left[\text{IN2d}\left[\begin{matrix}\text{Conv}\,(Z_k, ch_{\text{in}} = 2C_k\,, \\ ch_{\text{out}} = C_k, \text{groups} = C_k)\end{matrix}\right]\right]. \tag{2}$$

The mixing block also applies this tensor mixing strategy, and the generated U_e serves as the encoder's output and the decoder's processed input. The PW-MLP consists of multiple blocks, each utilizing a 1×1 convolution and a PReLU activation function, which allows nonzero gradients for negative inputs to mitigate the vanishing gradient problem. Finally, a tensor Y_k with dimensions $1 \times Y_k \times W_k$ is produced through one convolution. Unlike the PW-MLP in MAMB, the PW-MLP as a classifier uses Sigmoid as the activation function to ensure the results are between 0 and 1.

3) Upsampling Decoder: The decoder restores the spatial resolution through a series of upsampling layers, and each upsampling result is united with skip connection features to generate input for the subsequent layer. The input tensor of the kth layer of the decoder is reshaped through an upsampling operation to match the shape of Y_k. Let the upsampled tensor be U'_k, then U_k is defined as follows:

$$U_k = \text{PReLU}\left[\text{IN2d}\left[\text{Conv}\left(U'_k \odot Y_k\right)\right]\right] \tag{3}$$

where $\odot$ denotes the Hadamard product, representing the pixel-level skip connection attention mechanism. By multiplying with the mask Y_k, each pixel can be re-weighted to mitigate the false detection information produced by upsampling. Additionally, depthwise separable convolution [29] is used in the upsampling blocks.

4) Global Grouped Coordinate Attention: Conventional CNNs have the following limitations when dealing with complex change detection tasks: First, they are unable to process global information across the spatial dimensions of height

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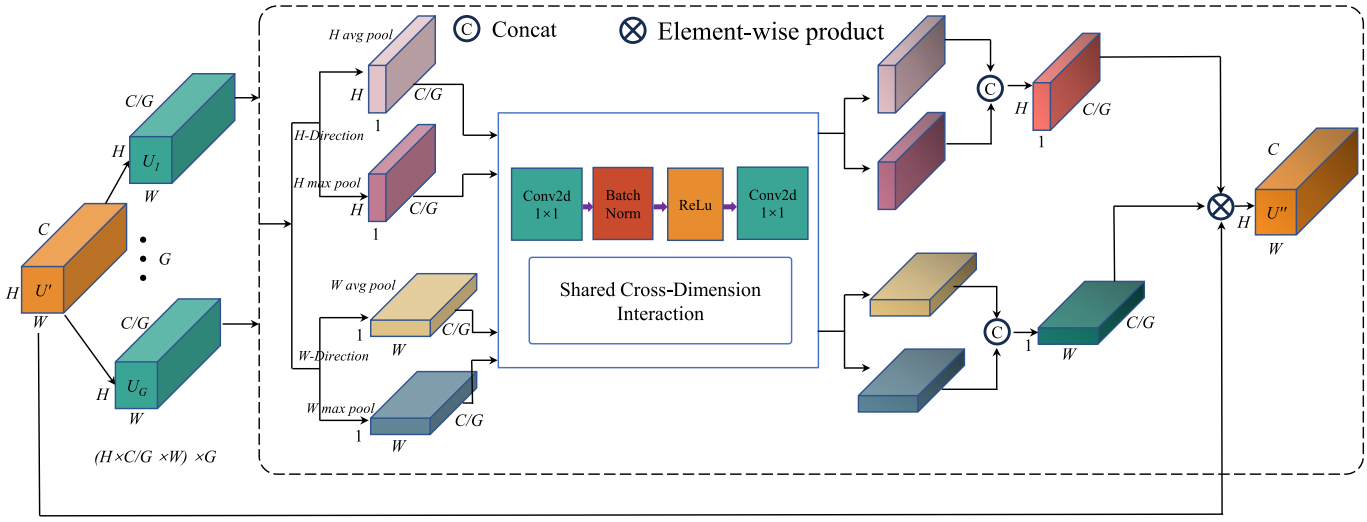


Fig. 6. GGCA structure.

and width in parallel, resulting in limited feature representation; Second, traditional convolution operations are unable to dynamically modify feature importance based on the feature responses at different channels and spatial locations, making it difficult to highlight key features; Finally, computational complexity and memory overhead are also critical issues when processing large-scale data. To surmount these barriers, this work suggests a lightweight attention mechanism called GGCA. The detailed structure is illustrated in Fig. 6 and consists primarily of the following components.

1) *Feature Grouping:* The input feature map is segmented into groups and processed by channels, minimizing the computational burden of each group while preserving the diversity and richness of feature representations. Assuming an input feature map U' with dimensions $R^{C\times H\times W}$ is divided into G subfeatures, the grouped feature map is denoted as follows:

$$U' \in R^{G\times \frac{C}{G}\times H\times W}. \tag{4}$$

2) *Global Pooling:* Next, global pooling operations, such as average and max-pooling, are applied to the grouped feature map to capture global information in multiple dimensions and refine the extensiveness of feature extraction

$$U'_{h,\text{avg}} = \text{AvgPool}\left(U'\right) \in R^{G\times \frac{C}{G}\times H\times 1} \tag{5}$$

$$U'_{h,\max} = \text{MaxPool}\left(U'\right) \in R^{G\times \frac{C}{G}\times H\times 1} \tag{6}$$

$$U'_{w,\text{avg}} = \text{AvgPool}\left(U'\right) \in R^{G\times \frac{C}{G}\times 1\times W} \tag{7}$$

$$U'_{w,\max} = \text{MaxPool}\left(U'\right) \in R^{G\times \frac{C}{G}\times 1\times W}. \tag{8}$$

3) *Shared Convolutional Block:* The shared convolutional block, including two 1 × 1 convolutions, a batch normalization layer, and a ReLU function, is implemented to process each grouped feature map while adjusting and restoring the channel dimension.

4) *Attention Weight Calculation:* The shared convolutional block's outputs are added together, then the Sigmoid is employed to produce attention weights in two directions, which are applied to weight the input feature map. This operation can enhance the representation of important features and suppress minor features. The calculation formula is as follows:

$$A_h = \sigma\left(\text{Conv}_s\left(U'_{h,\text{avg}}\right) + \text{Conv}_s\left(U'_{h,\max}\right)\right) \tag{9}$$

$$A_w = \sigma\left(\text{Conv}_s\left(U'_{w,\text{avg}}\right) + \text{Conv}_s\left(U'_{w,\max}\right)\right) \tag{10}$$

where σ represents the Sigmoid function and Convs represents the shared convolutional block.

5) *Applying Attention Weights:* The input feature map is adjusted based on the attention weights to generate the output feature map

$$U'' = U' \times A_h \times A_w \in R^{C\times H\times W}. \tag{11}$$

It should be noted that the attention weights Ah and Aw correspond to the input feature map's resolution, respectively.

5) Loss Function: Due to the highly imbalanced ratio of positive and negative samples in the forest change detection dataset (Table II), and the requirement for the model to focus more on identifying change regions in practical applications, we introduce the CE loss using the OHEM strategy and combine it with Dice loss to form a hybrid loss function to alleviate the impact of sample imbalance and enhance the model's capability to detect changes. Since the output feature maps are in the form of binary masks (0 and 1 represent no change and change, respectively), CE loss deals with the probability differences between the predicted results and the ground truth, enabling the model output to have a probabilistic interpretation. Dice loss, on the other hand, focuses more on evaluating the proportion of correctly classified positive samples to all positive samples [30], thereby mitigating the impact of disproportion between positive and negative samples. The detailed formulas for the aforementioned loss functions are as follows:

$$L_{\text{Dice}} = 1 - \frac{2G\cdot \text{Softmax}\left(P\right)}{G + \text{Softmax}\left(P\right)} \tag{12}$$

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TABLE III
COMPARISON RESULTS OF TINYFORESTCD WITH OTHER METHODS

Model	*Params(M)*	*FLOPs(G)*	*Pre (%)*	*Rc (%)*	*F1 (%)*	*IoU(%)*
FC-EF	1.35	3.57	70.27	67.42	68.82	52.46
FC-Siam-conc	1.55	5.32	67.72	71.25	69.44	53.19
FC-Siam-diff	1.35	4.72	68.83	66.71	66.75	51.23
IFNet	50.71	82.26	76.30	74.19	75.23	60.29
SNUNet	12.04	54.82	72.10	**87.40**	79.02	65.31
BIT	11.94	8.73	73.84	73.51	73.67	58.32
ChangeFormer	41.11	202.62	76.72	75.19	75.94	61.22
AMTNet	24.67	21.56	73.82	74.09	73.96	58.67
TinyCD	0.28	1.54	80.50	78.96	79.72	66.28
HANet	3.03	17.62	74.21	77.78	75.96	61.23
TinyForestCD(Ours)	0.29	1.54	**82.03**	79.31	**80.65**	**67.57**

$$L_{\mathrm{CE}}(p,y) = -\left[y\log(p) + (1-y)\log(1-p)\right] \tag{13}$$

$$L_{\mathrm{CE_OHEM}} = \mathrm{OHEM}(L_{\mathrm{CE}}) \tag{14}$$

$$L = L_{\mathrm{Dice}} + L_{\mathrm{CE_OHEM}} \tag{15}$$

where G represents the ground-truth label map, P represents the change map output by the model, p is the probability of the model predicting a sample as positive, and y is the true label value of the sample, which is either 0 or 1.

E. Model Comparison

To assess the effectiveness of the TinyForestCD model, the obtained results were compared and analyzed against other leading-edge open-source change detection frameworks. First, there are three pure CNN models: FC-EF, FC-Siam-Conc, and FC-Siam-Diff [11], which are all developed based on the U-Net structure. Next are the deeply supervised IFNet [12], the attention-based HANet [31], and SNUNet [13] models (using axial attention module and ensemble channel attention module, respectively), the Transformer-based BIT [16], ChangeFormer [17], and AMTNet [18]. Finally, there is TinyCD [20], which uses the Siamese U-Net architecture. TinyCD employs a global temporal and local spatial approach to extract low-level features, and experiments on high-resolution public datasets indicate that TinyCD surpasses the performance of most traditional convolutional models. It is important to note that all models were trained under identical conditions.

F. Evaluation Metrics and Experimental Environment

To verify the proposed model's effectiveness, we use four common evaluation metrics, defined as follows:

$$\mathrm{Pre} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FP}) \tag{16}$$

$$\mathrm{Rc} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN}) \tag{17}$$

$$\mathrm{F1} = \frac{1}{\mathrm{Pre}^{-1}+\mathrm{Rc}^{-1}} \tag{18}$$

$$\mathrm{IoU} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN}+\mathrm{FP}). \tag{19}$$

TP, TN, FP, and FN correspond to true positives, true negatives, false positives, and false negatives, respectively. Moreover, the computational complexity of the algorithms is assessed based on the number of parameters (Params) and floating-point operations (FLOPs).

In this experiment, Python 3.10 is used as the programming language, and computations are performed on an Intel i7-12700F CPU and an NVIDIA RTX 3060 GPU. The selected deep learning framework is PyTorch 2.0.1. To maintain experiment reproducibility, a fixed random seed is applied. The Adam optimizer is configured with an initial learning rate of 3×10^{-3} and a weight decay coefficient of 1×10^{-2}. The OHEM sample filtering threshold is set to 0.7 based on experience. In terms of data augmentation, random horizontal and vertical flips, as well as random rotations within a certain range, are applied as geometric transformations to simulate different imaging angles and directions, improving the model's rotational and flip invariance. Furthermore, nongeometric augmentation techniques such as Gaussian blur, random adjustments in brightness and contrast, are applied to mimic noise and illumination variations during the imaging process, reinforcing the model's stability under variations in image quality. Through appropriate data augmentation strategies, we can effectively expand the diversity within the training dataset and optimize the model's capacity for generalization and adaptability to complex scenes. Given the restrictions of GPU memory and processing power, the batch size is configured as 4, and training is conducted for 100 epochs. Only the best-performing model weights are saved as the final result.

III. RESULTS

A. Comparison With Leading-Edge Methods

Table III presents the accuracy metrics and model complexity results for all models evaluated. From the table, the TinyForestCD model is a lightweight model with low computational overhead. Moreover, TinyForestCD delivers superior performance in key metrics, including $F1$-score, precision, and IoU. Compared to TinyCD, the precision, recall, $F1$-score, and IoU increase by 1.53%, 0.35%, 0.93%, and 1.29%, respectively. Fig. 7 shows the visualization comparison results of randomly selected forest change test sets, which can intuitively compare the differences between each model. Different colors and a yellow box are used in the figure to display the detailed results of the models, where yellow and red represent missed detections and false detections, respectively, and green represents the correctly detected parts by the models.

From Fig. 7, TinyForestCD demonstrates superior performance in reducing missed detections. Specifically, in (a) and (c), FC-EF, FC-Siam-conc, and FC-Siam-diff exhibit significant detection omissions; in (b) and (c), IFNet and

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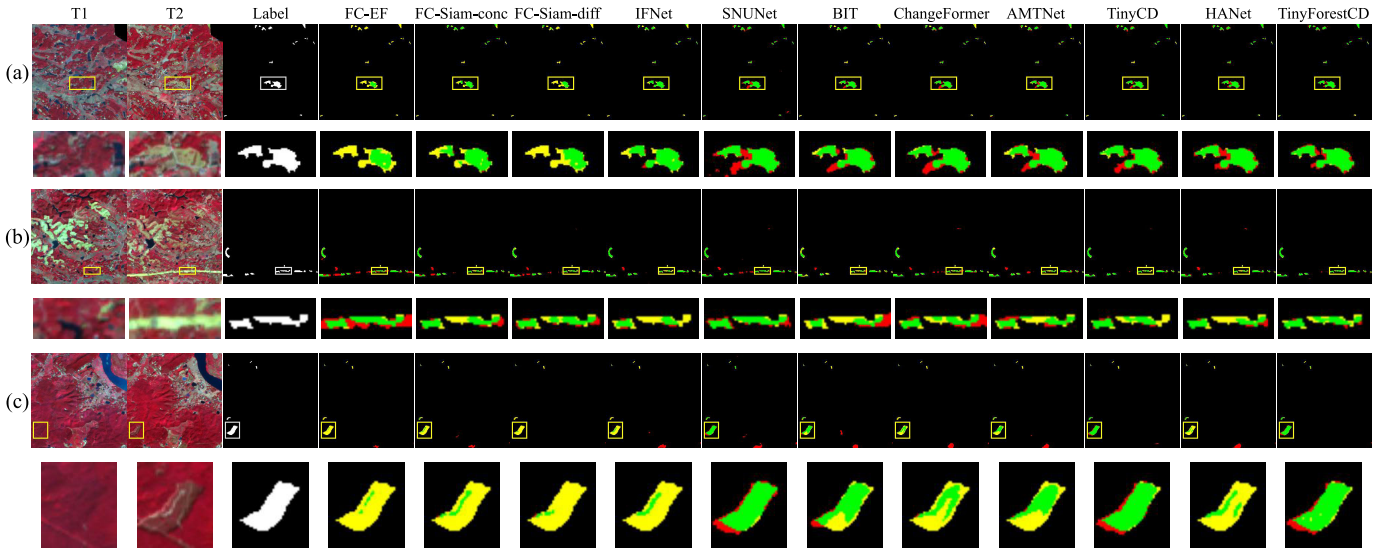


Fig. 7. Visualization results of different models on the test set, where (a)–(c) are zoomed-in examples of obvious forest change regions within the study area. In the figure, yellow and red represent missed detections and false detections, respectively, and green represents the parts correctly detected by the model.

TABLE IV
RESULTS OF THE ABLATION STUDY

Model	*Params(M)*	*FLOPs(G)*	*Pre (%)*	*Rc (%)*	*F1 (%)*	*IoU(%)*
Base	0.28	1.54	80.50	78.96	79.72	66.28
Base-Hybrid Loss	0.28	1.54	78.54	**81.34**	79.91	66.55
Base-GGCA	0.29	1.54	**82.45**	78.60	80.48	67.33
TinyForestCD(Ours)	0.29	1.54	82.03	79.31	**80.65**	**67.57**

HANet have relatively serious missed detections. Compared to BIT, ChangeFormer, AMTNet, and TinyCD, TinyForestCD has fewer yellow pixels. Second, our model achieves a lower false detection rate. In (a), SNUNet, BIT, ChangeFormer, TinyCD, HANet, and, in (b), FC-EF, FC-Siam-conc, FC-Siam-diff, SNUNet, BIT, ChangeFormer, and AMTNet clearly have more red pixels, indicating that these models have more false detection areas. By observing the red pixel regions, it can be found that most of these regions are nonforest vegetation, suggesting that other models have difficulty distinguishing between forest and nonforest areas, leading to false detections. Therefore, TinyForestCD performs better than other models in suppressing false detections.

B. Ablation Study

1) Ablation Study of the Proposed Model: TinyForestCD is mainly composed of the TinyCD baseline with the addition of GGCA and an improved loss function. Therefore, to separately assess the effect of the hybrid loss function and GGCA, an ablation study is conducted to compare four models: TinyCD (Base), TinyCD with the improved hybrid loss function, TinyCD with the addition of GGCA, and TinyForestCD. The evaluation metric results are outlined in Table IV, while the visualization outcomes are illustrated in Fig. 8.

Table IV shows that using the improved hybrid loss function, the $F1$-score and IoU increase by 0.19% and 0.27%, respectively, when benchmarked against the baseline. After adding the GGCA module, the $F1$-score and IoU increase by 0.76% and 1.05%, respectively, compared to the baseline model. The TinyForestCD model, which combines the hybrid loss function and the GGCA module, achieves improvements of 0.93% in $F1$-score and 1.29% in IoU compared to the baseline model. From Fig. 8(a)–(c), TinyForestCD has fewer accumulated yellow and red pixels compared to other models, indicating that the model can better balance missed detections and false detections.

2) Ablation Study of Different Numbers of GGCA Groups: From Table IV, we can see that after adding the GGCA module alone to the baseline, the recall rate decreases to a certain extent. Therefore, using the GGCA module alone may over-suppress irrelevant information, resulting in a certain degree of missed detection. Therefore, we explored the changes in precision and recall under different numbers of groups G, to balance the strength of the GGCA module. As shown in Table V, increasing the number of groups G leads to higher recall but lower precision. This is because a greater number of groups allows for more fine-grained feature extraction, improving recall, but due to the reduction in the number of channels per group, the model becomes more prone to misclassification, which lowers precision. Our experimental results indicate that the optimal number of groups is 4, as it achieves the best balance between precision and recall. To further clarify the sensitivity of the grouping parameter under different conditions, we conducted additional comparative experiments using the Shaoguan area. The results show that the model's performance remains stable across different values of G, with the best performance still achieved at $G = 4$.

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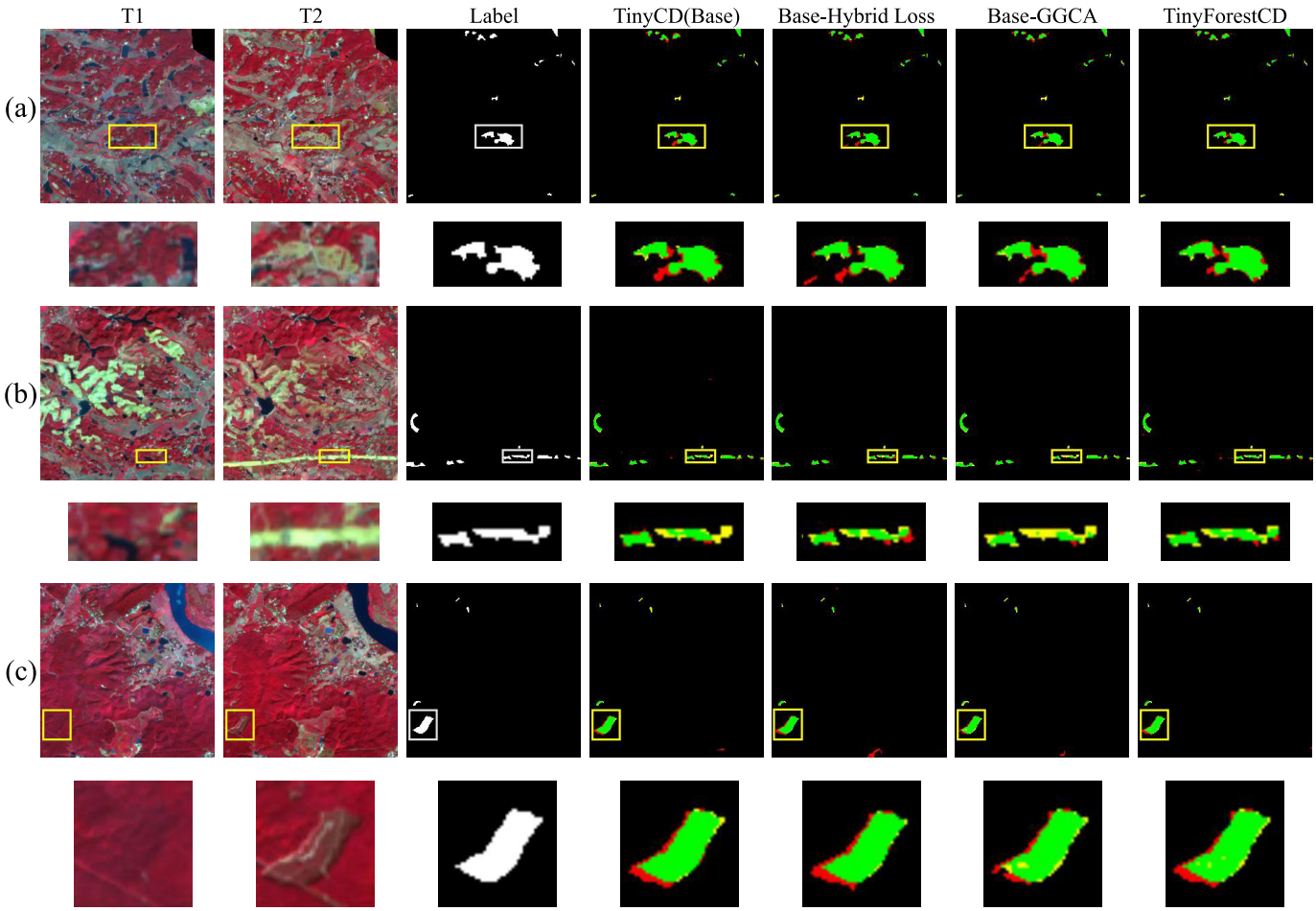


Fig. 8. Visualization results of the ablation study, where (a)–(c) are the same three zoomed-in examples of obvious forest change regions as Fig. 7. In the figure, yellow and red represent missed detections and false detections, respectively, and green represents the parts correctly detected by the model.

TABLE V
ABLATION STUDY OF DIFFERENT GROUP NUMBERS IN GGCA

Group	Study area				Shaoguan area			
	Precision (%)	*Recall (%)*	*F1 Score (%)*	*IoU (%)*	*Precision (%)*	*Recall (%)*	*F1 Score (%)*	*IoU (%)*
2	**83.47**	76.31	79.73	66.29	**82.19**	78.83	80.47	67.33
4	82.45	78.60	**80.48**	**67.33**	81.15	81.13	**81.14**	**68.27**
8	79.91	80.46	80.18	66.92	78.57	82.97	80.71	67.66
16	78.65	**81.25**	79.93	66.57	77.33	**83.76**	80.42	67.25

TABLE VI
EVALUATION RESULTS OF GGCA SUPERIORITY

Model	*Params(M)*	*FLOPs(G)*	*Pre (%)*	*Rc (%)*	*F1 (%)*	*IoU(%)*
Base	0.28	1.54	80.50	78.96	79.72	66.28
Base-scSE	0.29	1.54	81.84	77.92	79.83	66.43
Base-CA	0.29	1.54	80.97	79.18	80.06	66.76
Base-CCA	0.29	1.62	81.31	78.81	80.04	66.72
TinyForestCD(GGCA)	0.29	1.54	**82.03**	**79.31**	**80.65**	**67.57**

C. GGCA Superiority Assessment

To validate the superiority of GGCA, the study also compares it with some widely used lightweight attention mechanisms, including scSE [32], CoordAttention (CA) [33], and CrissCrossAttention (CCA) [34]. It should be noted that, except for the attention mechanism, all other parameters remain the same. The evaluation metric results are outlined in Table VI, while the visualization outcomes are illustrated in Fig. 9.

Table VI shows that incorporating scSE, CA, CCA, and GGCA leads to $F1$-score improvements of 0.11%, 0.34%,

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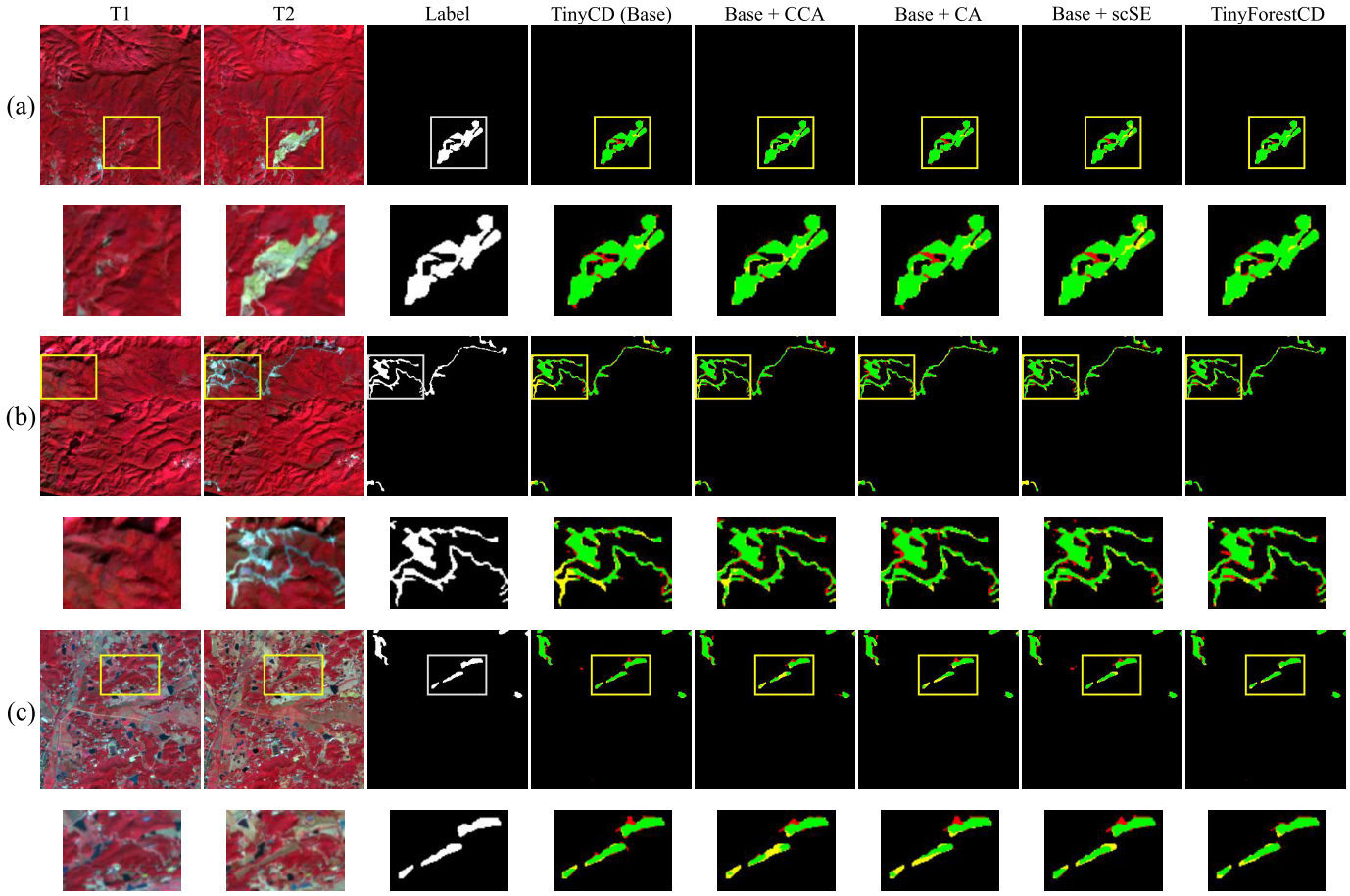


Fig. 9. Visualization results of GGCA superiority assessment, where (a)–(c) are zoomed-in examples of obvious forest change regions within the study area. In the figure, yellow and red represent missed detections and false detections, respectively, and green represents the parts correctly detected by the model.

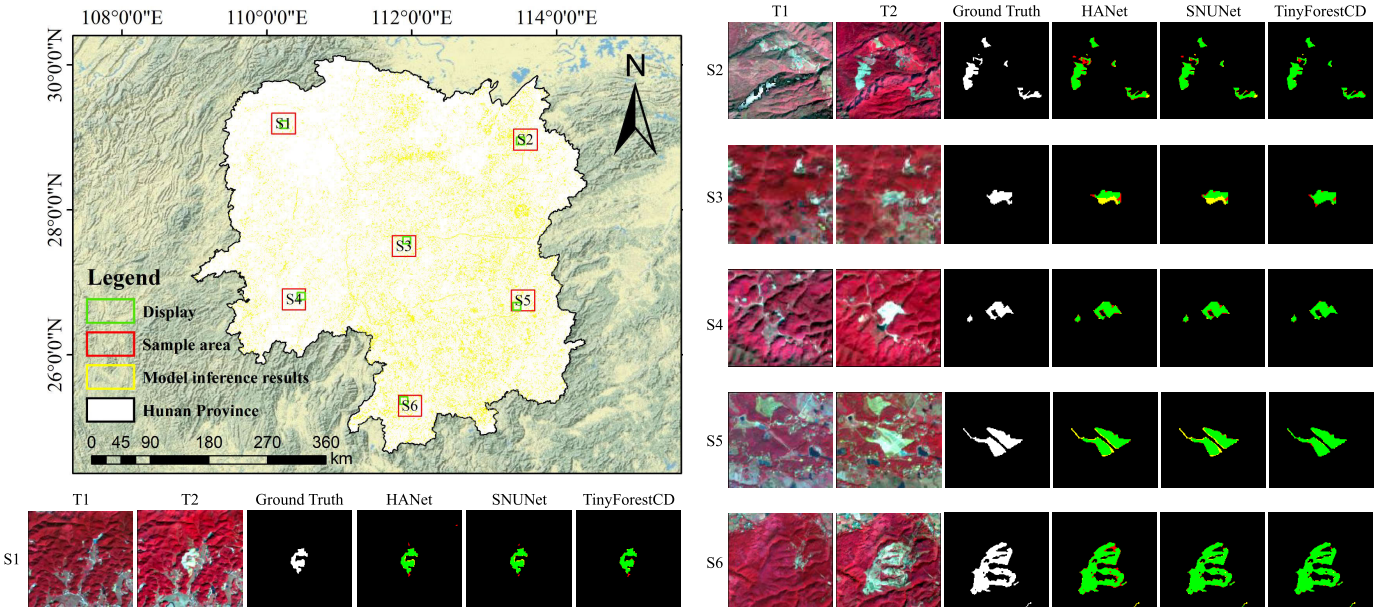


Fig. 10. Hunan Province deployment results display. In the figure, yellow and red represent missed detections and false detections, respectively, and green represents the parts correctly detected by the model.

0.32%, and 0.93%, respectively, over the baseline model, while IoU increases by 0.15%, 0.48%, 0.44%, and 1.29%, respectively. Moreover, the TinyForestCD model outperforms the models with other attention mechanisms in all four metrics:

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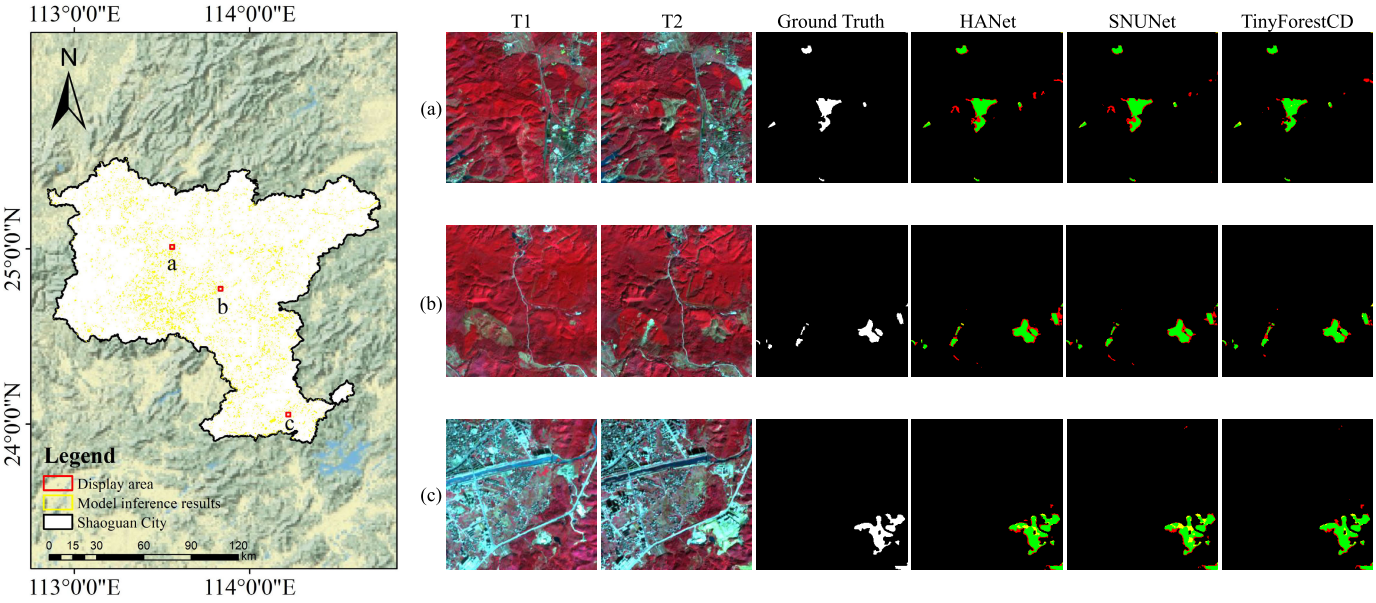


Fig. 11. Shaoguan City deployment results display, where (a)–(c) are zoomed-in examples of obvious forest change regions within the Shaoguan City. In the figure, yellow and red represent missed detections and false detections, respectively, and green represents the parts correctly detected by the model.

TABLE VII

ACCURACY AND EFFICIENCY EVALUATION OF CROSS-REGIONAL DEPLOYMENT RESULTS FOR DIFFERENT MODELS

Model	Precision indicators	S1	S2	S3	S4	S5	S6	Shaoguan area
TinyForestCD	Pre/%	83.95	81.87	80.92	76.12	85.34	81.29	74.68
	Rc/%	81.05	74.18	72.64	82.45	84.78	84.56	82.02
	F1/%	82.47	77.85	76.62	79.15	85.05	82.89	78.18
	IoU/%	70.23	63.84	62.19	65.78	74.01	70.82	64.17
	FPS	72.13	70.19	71.72	68.72	68.20	68.87	68.72
HANet	Pre/%	71.78	77.96	72.76	74.04	82.60	65.55	65.17
	Rc/%	69.28	74.88	66.30	83.14	80.02	79.13	87.26
	F1/%	70.51	76.39	69.38	78.33	81.29	76.89	74.63
	IoU/%	54.45	61.88	53.12	64.37	68.48	55.89	59.52
	FPS	36.67	36.38	35.59	37.64	38.02	36.42	34.68
SNUNet	Pre/%	82.86	86.63	66.27	70.71	86.25	70.45	67.92
	Rc/%	73.38	76.11	70.11	83.81	78.16	76.76	80.11
	F1/%	77.84	81.13	68.13	76.01	79.34	73.54	73.51
	IoU/%	63.72	68.11	51.67	62.21	69.50	58.06	58.11
	FPS	38.54	39.64	40.15	40.70	39.93	38.76	37.14

precision, recall, $F1$, and IoU, highlighting the superiority of our proposed attention mechanism. Additionally, Fig. 9 displays a comparison of the visualization outcomes of these models. From (a), it can be observed that in the obvious forest change regions, the detection outcomes of each model show no significant differences. In the change regions with more complex shapes, as shown in (b), TinyCD exhibits obvious missed detections. After adding the CCA module, missed detections are reduced to some extent, but the phenomenon of missed detections is still relatively evident. In (c), compared to other models, our model exhibits the lowest number of falsely detected pixels. Therefore, the TinyForestCD model using GGCA has certain superiority in suppressing false detections.

D. Model Cross-Region Transfer Learning and Analysis

To assess the proposed method's effectiveness, we used TinyForestCD and two other better models, SNUNet and HANet, to conduct actual deployments in the entire Hunan Province and Shaoguan City, resulting in a vector map illustrating forest reduction areas in the region between 2020 and 2022. The experiment randomly selected six sample areas for quantitative analysis, each with an area of 1024 km^2 were randomly selected in Hunan Province, and all areas of Shaoguan City participated in the evaluation. In terms of efficiency evaluation, for each region and each model, we performed five independent runs and reported the average FPS as the final result. All models were evaluated in pure inference mode, with the same input size and the same pre/postprocessing; timing does not include data loading and I/O. The accuracy evaluation and efficiency evaluation results of the six sample plots in Hunan Province and Shaoguan City are shown in Table VII, and some areas are randomly selected for display (Figs. 10 and 11).

In the deployment results of Hunan Province, the S5 region has the highest accuracy, with an $F1$ of 85.47%; the S3 region has the lowest accuracy, with an $F1$ of 77.05%. The average

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metrics of the six sample areas (precision 82.00%, recall 80.31%, $F1$-score 81.06%, and IoU 68.25%) do not differ significantly from the test results in the dataset, indicating that the TinyForestCD model can still maintain stable performance overall when performing large-scale forest change detection tasks in practice. In terms of adaptability to different scenarios, the S5 sample area exhibits high detection accuracy, while the accuracy in the S3 region is relatively low. Analysis reveals that the S5 sample area partially overlaps with the Changsha–Zhuzhou–Xiangtan area, so a small portion of the data is consistent with the training dataset, resulting in higher detection accuracy in the S5 sample area. Due to the dense distribution of farmland and fragmented plots in the S3 sample area, the spectral features of the objects are relatively close to those of forests, increasing the classification difficulty and leading to the model deployment accuracy being lower than the dataset accuracy to a certain extent. Overall, the accuracy differences between sample areas are mainly influenced by three types of factors: 1) differences in the spatial distribution characteristics and attributes of land cover within different sample areas, such as the S1 sample area being mostly forest and the S3 sample area being mostly farmland; 2) shadow effects caused by terrain undulations, which can easily cause missed detections in mountainous areas; and 3) the phenomenon of identical objects exhibiting varying spectra and distinct objects displaying similar spectra, resulting from changes in illumination conditions.

In the results of the deployment in Shaoguan City, TinyForestCD achieved the best overall performance (F1 = 78.18%, IoU = 64.17%) with the lowest number of parameters (0.29 M), verifying its advantages of efficient parameter utilization and task adaptability. HANet's high recall rate (87.26%) indicates that it is more inclined to detect potential change areas, but its lower precision rate (65.17%) means that there are more false positives. This feature may be suitable for scenarios that are sensitive to missed detections. The large-parameter model SNUNet performed better than HANet in the dataset, but its performance was lower than HANet in the Shaoguan area, so there may be overfitting.

In terms of inference speed, TinyForestCD generally achieves the highest frame rates across various regions, with relatively stable performance; HANet's frame rates range from 34.68 to 38.02 across regions, and its inference speed is the slowest in every area. Although SNUNet offers an overall improvement in inference speed compared to HANet, it still lags significantly behind TinyForestCD. Therefore, TinyForestCD is the most computationally efficient and can handle higher frame rates. This is particularly crucial when deploying models in real-time applications or when hardware resources are limited.

IV. Discussion

A. Impact of the Hybrid Loss Function and the GGCA Module on the Model

The ablation study results reveal that incorporating the hybrid loss function and the GGCA module significantly enhances model performance, though their mechanisms of improvement vary. The hybrid loss function can significantly increase the recall rate, especially in capturing subtle changes, by directly optimizing the overlap between the predicted regions and the true change regions. On the other hand, the GGCA module effectively improves the model's precision by fusing multidimensional global information and enhancing the weights of key features. By combining the two, the model finds an optimal tradeoff between precision and recall, validating the importance of their collaborative optimization in change detection tasks.

B. Analysis of Factors Suppressing False Detections in the Model

The GGCA module in this study demonstrates significant advantages in suppressing false detections compared to other lightweight attention mechanisms. Fusing global information and attention mechanisms, it enables the model to dynamically focus on target regions, enhancing its ability to discriminate features in complex scenes. Furthermore, the strategy of combining the hybrid loss function with OHEM enables the model to concentrate efficiently on critical change regions, thereby reducing false detections. Experiments show that the TinyForestCD model significantly reduces the interference of seasonal vegetation changes (Fig. 12 left) and nonforest vegetation changes (Fig. 12 right) on detection results, performing exceptionally well in complex change scenarios and becoming the best-performing method in all comparative experiments.

Additionally, to ensure that the improvement in the model was not due to random variation, we performed 5000 iterations of bootstrap resampling and found that the difference in IoU between the TinyForestCD model and the TinyCD model was 1.30%, with a 95% confidence interval of [0.25%, 2.53%] (Fig. 13). Since the confidence interval does not include 0, we can reasonably conclude that the observed difference is statistically significant.

C. Model Actual Deployment Analysis

The lightweight design of TinyForestCD enables it to run stably on ordinary hardware (such as Intel i7-12700F CPU and NVIDIA RTX3060 GPU) without relying on high-end computing equipment, greatly reducing the hardware threshold for actual deployment. This characteristic makes it particularly suitable for scenarios that require rapid response, such as forest fire monitoring and disaster assessment. In the large-scale deployment experiment in Hunan Province, the model demonstrated high accuracy and stability when dealing with complex land feature changes.

In theory, the smallest detectable change in Sentinel-2 corresponds to a single pixel (100 square meters), which means the model can detect changes at the pixel level. However, compared to higher-resolution sensors (such as PlanetScope or GaoFen-2), Sentinel-2 has relatively coarse resolution, leading to greater spatial averaging and positional jitter in boundary delineation. Therefore, when mapping small or fragmented changes, results based on Sentinel-2 may exhibit slightly larger fluctuations, as finer resolution data can capture changes with higher geometric accuracy. Future research can further

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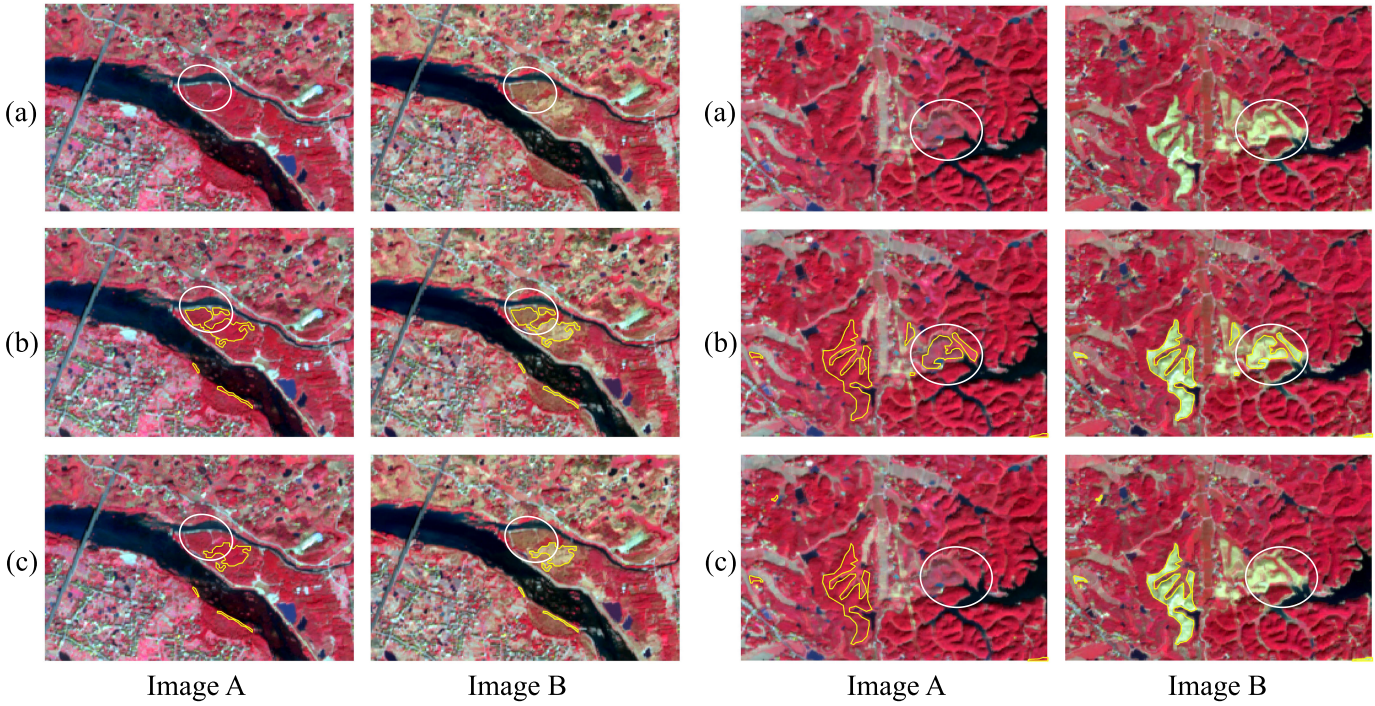


Fig. 12. Examples of false detections in forest change. (a) Original image. (b) TinyCD prediction result. (c) TinyForestCD prediction result. The yellow areas are the prediction results of TinyCD and TinyForestCD, and the white circles indicate false detection regions. The white circle in the left region of the figure represents false detections caused by seasonal changes in forests, while the white circle in the right region indicates that the changes detected by the model are nonforest vegetation.

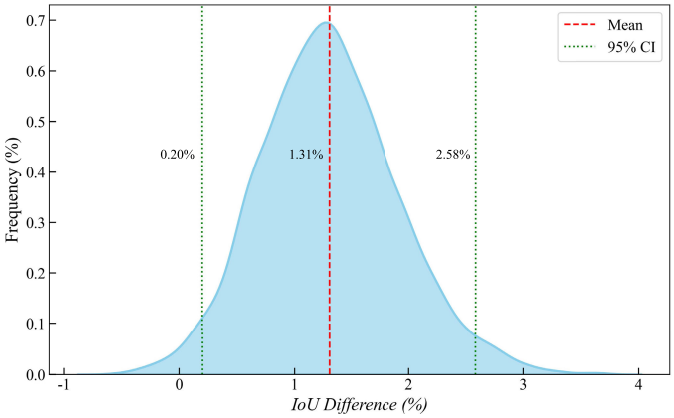


Fig. 13. Significance test results.

improve the model's environmental adaptability by introducing high-resolution remote sensing imagery, enriching training samples, and integrating terrain auxiliary information. Furthermore, cross-region deployment validation for different areas will help enhance the model's cross-regional applicability and generalizability.

D. Limitations and Future Prospects

In this study, only conventional metrics such as the $F1$-score were used to evaluate change detection performance. However, we also recognize the potential value of other scientific evaluation systems, such as the evaluation method proposed by Olofsson et al. [35]. This method is suitable for the accuracy evaluation of large-scale forest change detection tasks, so we plan to explore its use in future work. By adopting two evaluation methods, we can get a more comprehensive understanding of the performance of the model and ensure that we capture accuracy from multiple perspectives. This will help to further enhance the robustness of our model validation process.

Second, although the TinyForestCD model demonstrates strong performance in detecting areas of forest loss, and the use of hybrid loss functions combined with OHEM has shown effectiveness in addressing class imbalance, we acknowledge that the applicability and potential limitations of this approach across different types of forest changes have not been fully explored. Forest change detection involves more than identifying whether a change has occurred—it also requires distinguishing the specific type and underlying driver of the change. Forest loss can result from a variety of factors, including anthropogenic disturbances (e.g., illegal logging, infrastructure development, and fire clearance) and natural causes (e.g., pest outbreaks, landslides, and windstorms), each exhibiting distinct spatial patterns, scales, and spectral characteristics. The current loss function primarily optimizes performance for binary change/no-change classification. However, as pointed out in related works [36], OHEM also has certain limitations. For example, over-emphasizing "difficult examples" in the early stages of training may cause the model to fall into a local optimum, especially when the detector does not yet have good discrimination ability.

In addition, a quantitative comparison of false positive categories would provide more insight into the effectiveness of

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the GGCA mechanism. However, due to the lack of pixel-level reference data distinguishing specific sources of false positives (e.g., seasonal vegetation phenology and crop rotation) across the entire study area, a category-level statistical breakdown of false positives is not available in this version. A fine-grained reference dataset will be developed in future research to enable systematic, category-based evaluation of false positives in future work.

In addition to the approaches mentioned above, future research can enhance the model's sensitivity and discrimination ability to multiple change types by integrating multisource remote sensing data, hierarchical attention mechanisms, and domain adaptation techniques. Moreover, refining the training sample selection—potentially through class-aware hard example mining strategies—may further improve the model's generalization in complex forest change scenarios. Accurate identification and classification of change types will not only enable a more comprehensive understanding of the spatial patterns of forest dynamics but also provide critical insights into their socio-economic and environmental drivers, thereby supporting more targeted and effective forest conservation and restoration policies.

V. Conclusion

This work introduces a lightweight model for detecting forest changes, named TinyForestCD, which combines EfficientNet_b4 as the backbone network and incorporates a MAMB, a Mixing Block, and a GGCA module. Through multilevel feature enhancement and attention mechanisms, the model effectively extracts key features in forest changes while suppressing irrelevant interference. To address the challenge of highly imbalanced forest change data, the research utilizes a hybrid loss function that combines CE loss and Dice loss, along with an OHEM strategy, to enhance the model's detection precision and robustness. Through various comparative experiments, the findings indicate that the proposed forest change detection model, TinyForestCD, surpasses other models across critical metrics, achieving 82.03% precision, 79.31% recall, 80.65% $F1$-score, and 67.57% IoU. Therefore, TinyForestCD has a better ability to suppress false detection than other models in extremely unbalanced forest change detection datasets. Moreover, the optimized model architecture enables it to have only 0.29 M parameters and 1.54G FLOPs, showcasing its lightweight advantage and suitability for large-scale applications. The actual deployment experiment shows that the model achieves an average $F1$ metric of 81.06% and an IoU performance of 68.25% in the sample areas of Hunan Province, consistent with the test set results. In addition, in the deployment results of the Shaoguan area, the model also showed the best performance compared with other models, validating the model's stability and adaptability. Overall, TinyForestCD demonstrates excellent performance in avoiding missed detections and reducing false detections, exhibiting the potential for efficient and accurate large-scale forest change detection. Offering dependable technical assistance for the real-time monitoring and effective management of forest resources.

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